

SPACE TECH: THE ROAD TO BHARATIYA ANTARIKSH STATION (2026-2035)



1. GAGANYAAN-1 (G1) | MARCH 2026



- **Uncrewed Human-Rated Orbital Test Flight**
- **Humanoid (Vyommitra):** Female half-humanoid, monitors ECLSS, simulates human functions, tests re-entry
- **Launch Vehicle:** HLVM3 (Human-Rated LVM3)
- **Objective:** Validate safety-critical systems



3. BHARATIYA ANTARIKSH STATION (BAS) | GOAL 2035

- **Phase 1 (BAS-1):** Target launch by 2028 (~10 tonnes)
- **Full Vision:** Multi-modular station in LEO (~400 km)
- **Key Features:** Indigenous life support, Bharat Docking System (BDS), Microgravity labs



2. SPADEX: THE DOCKING MILESTONE (JAN 16, 2025)

- **Space Docking Experiment**
- **Achievement:** Autonomous docking of 'Chaser' & 'Target' satellites
- **Foundational Tech for:** Bharatiya Antariksh Station (BAS), Chandrayaan-4 sample return, In-Orbit Refueling



4. BACKING POLICY: INDIAN SPACE POLICY 2023

- **Shift in Role:** ISRO moves to R&D hub
- **Private Sector Engagement:** Allows launch rockets, own satellites (Non-Governmental Entities)
- **IN-SPACE:** Single-window regulator & promoter
- **Objective:** Increase global space economy share to 10% by 2030

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KEY TAKEAWAY

- **Technological Linkage:** SpaDeX is the prerequisite for BAS-1 (2028), which is the stepping stone for Human Lunar Landing by 2040



ISRO'S STRATEGIC FRONTIERS: THE LUNAR PATHWAY & HUMAN FLIGHT

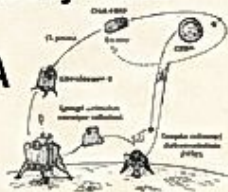
GS PAPER III - SCIENCE & TECHNOLOGY | PRELIMS 2026 ANCHOR

1. CHANDRAYAAN-4 & LUPEX

 **MISSION GOAL:** Collect ~3 kg of lunar regolith and return it to Earth from the Moon's South Pole (~84°–86° S).

 **ARCHITECTURE:** A complex 5-module mission involving Ascender, Descender, Transfer, Re-entry, Propulsion, utilizing multiple LVM3 launches.

- **LUPEX (CHANDRAYAAN-5):** Joint ISRO-JAXA collaboration aiming to survey lunar water ice quality and quantity. ISRO provides the Lander, JAXA provides the H3 Rocket and 350 kg Rover.



- **TARGET AREA:** Permanently Shadowed Regions (PSRs) at the lunar poles.

2. AXIOM MISSION 4 (AX-4) & GAGANYAAN

 **INDIAN ASTRONAUT:** Group Captain Shubhanshu Shukla flew to the ISS in July 2025 aboard SpaceX Dragon Grace.

 **MISSION DURATION:** 18 days on the International Space Station, serving as critical prep for India's human spaceflight.

 **KEY OBJECTIVES:** Real-time experience in health monitoring, spacecraft operation, crew coordination. Over 60 scientific experiments conducted.

 **STRATEGIC VALUE:** Accelerates timeline for Gaganyaan's first crewed mission, enhancing technical and operational capabilities for India's own space lab and astronaut corps.



3. NISAR: GLOBAL RADAR SURVEY

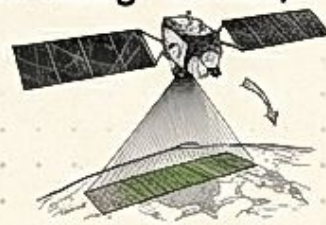
 **FULL NAME:** NASA-ISRO Synthetic Aperture Radar, a collaborative Earth-observing satellite.

 **WORLD FIRST:** First dual-frequency SAR satellite (using L-band and S-band radar).

 **CAPABILITIES:** Maps entire Earth's surface every 12 days; 1 cm accuracy in measuring land change.

 **PRIMARY APPLICATIONS:** Monitoring ice sheets, volcanic activity, earthquakes, coastal erosion, forests, and agriculture.

 **RELEVANCE FOR INDIA:** Provides accurate data for disaster management, natural resource mapping, climate change studies, and urban planning.



INSIGHT BOX: PRELIMS HIGH-YIELD POINTS

UPSC PRELIMS ANGLE: Chandrayaan-4 (Sample Return), LUPEX (Water Ice Search).

COMMON TRAP: Chandrayaan-4 returns samples; don't confuse with Chandrayaan-3. NISAR is a 'dual-frequency radar' (L-band & S-band) and maps every 12 days. AX-4: Provided ready experiences for Gaganyaan crews, focused on microgravity physiology. For NISAR, focus on 'crustal deformation' mapping.

GAGANYAAN: First human flight scheduled post-Ax-4.

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by Abhishek Pathak

EMPOWERING INDIA: EMERGING TECHNOLOGIES for UPSC PRELIMS 2026

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INDIA AI MISSION

Implementation of the Seven Pillars

AIKosh:

- Large Datasets:



High-Quality,
Non-Personal Datasets

Establishment of GPU Clusters



Over 38,000 GPUs
High-Performance Computing

Sovereign AI Models



BharatGen



Sarvam AI

Indigenous Development
Indian Languages & Context

Digital India

NATIONAL SUPERCOMPUTING MISSION

Building Computational Power

Total Capacity



39 Petaflops Capacity
Across 37 Locations

Indigenous Infrastructure



Rudra Server:
Native Hardware

Software Stacks



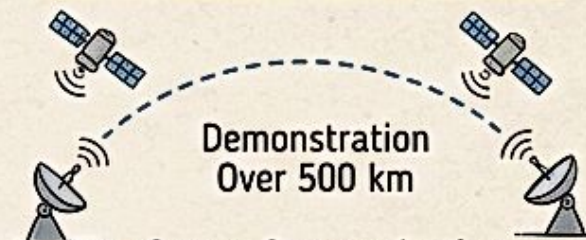
Indigenous
Software Stacks
Atmanirbhar
Computing

- Indigenous Software Stacks
- Atmanirbhar Computing

NATIONAL QUANTUM MISSION (NQM)

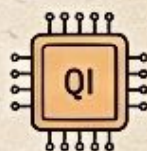
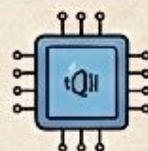
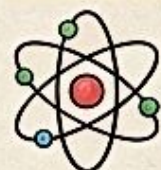
Key Achievements 2025-2026

Quantum Key Distribution (QKD)



- Secure Communication

Indigenous Quantum Processors



- 50-100 Qubit Processors
Target: Drug Discovery

Applications:

Medicine, Material Science

GENERATIVE AI GOVERNANCE

Ethical Frameworks & Leadership

- New Delhi Declaration on AI Impact (2026)



- Global Accord:
Bias Mitigation,
Responsible AI

- India AI Impact Summit



- Focus:
Addressing Bias,
Accountable AI

- Ethical AI
Global Standards



HEALTH & BIOTECHNOLOGY FOR UPSC PRELIMS 2026



GENOMICS: 10,000 GENOME PROJECT & CLINICAL GENOMICS

* Genome India Project (GIP) Milestone



Seqs 10,000 whole genomes, representing 83 diverse ethno-linguistic groups
Key Hub: Data in Indian Biological Data Centre (IBDC).

* Clinical Genomics in Public Hospitals



Focus: Precision Medicine in Oncology
Targeted treatments for cancer based on genetics (e.g., BRCA1/2 mutations).
Shift from "one-size-fits-all"



ONE HEALTH APPROACH

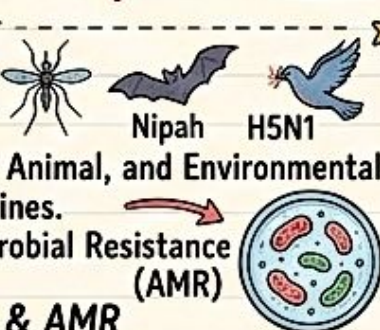
* Theme & Focus

Interconnected Human, Animal, and Environmental Health. Bridging disciplines.
Specifically for Antimicrobial Resistance (AMR)



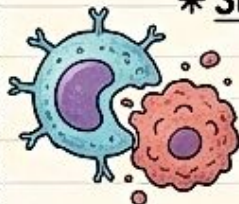
* Viral Surveillance & AMR

Focus: Pathogens like Nipah, Zika, H5N1 (Avian Flu)
AMR: Monitoring superbugs, drug misuse in livestock.
Goal: Prevention, early detection.



BREAKTHROUGH THERAPIES: CAR T-CELL & mRNA/DNA

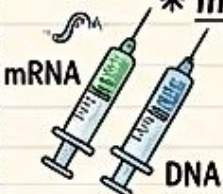
* Scaling CAR T-Cell Therapy



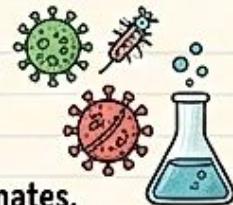
Gene-engineered T-cells to attack cancer cells
Indian context: Reduced cost, indigenous options (e.g., NexCAR19)
Application: specific types of cancer (leukemia, etc.)



* mRNA/DNA-Based Therapeutics



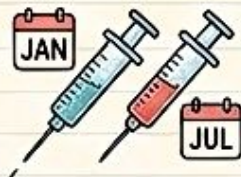
mRNA platform (proven in COVID) for Tropical Diseases (Dengue, Malaria)
DNA vaccines (like Zycov-D)
Potentially more stable in tropical climates.



LONG-ACTING HIV PREVENTION

* Clinical Breakthroughs

Focus: Prevention of HIV, not just treatment.
Shift from daily pills to rare injections.

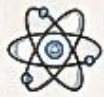


* Mechanism & Injections

Drug type: Potentially Lenacapavir (like")
Mechanism: Bi-annual (every 6 months) subcutaneous injections.
Goals: Complete HIV prevention, better compliance.



CRITICAL TOPICS FOR UPSC PRELIMS 2026: DEFENCE & NUCLEAR TECHNOLOGY

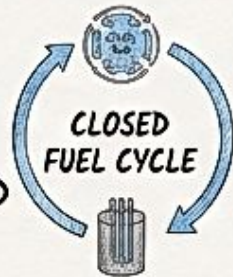


INDIA'S 3-STAGE NUCLEAR PROGRAMME



STAGE 2 ACHIEVED (APRIL 2026)

PROTOTYPE FAST BREEDER REACTOR (PFBR) at KALPAKKAM



KEY BREAKTHROUGH:
FIRST CRITICALITY REACHED

- FUEL: MOX (Mixed Oxide)
- COOLANT: Liquid Sodium
- BREEDS: Plutonium-239 & Uranium-233 from Thorium-232



SMALL MODULAR REACTORS (SMRs)

↳ <300 MW

- BENEFITS: Scalable, Safer, Shorter Construction
- DEVELOPMENT: Indian SMR design (220 MWe, PHWR tech-based)

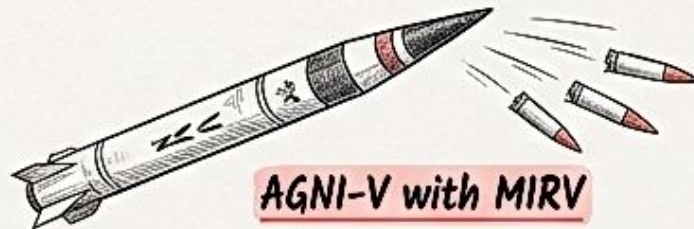


MISSILE SYSTEMS & STRATEGIC DETERRENCE



K-6 SLBM

- RANGE: ~8,000 km (Intercontinental)
- PLATFORM: Future SS-class SSBNs
- STATUS: Under Advanced Development/Testing



AGNI-V with MIRV

- MIRV: Multiple Independently Targetable Re-entry Vehicle
- MISSION: "DIVYASTRA" (March 2024 success)
- RANGE: 5,000+ km
- SIGNIFICANCE: Counter-countermeasures, Enhanced Deterrence

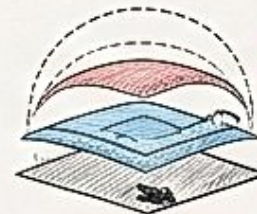


AEROSPACE & AIR DEFENCE INDIGENISATION



AMCA (Advanced Medium Combat Aircraft)

- TYPE: 5th-Generation Stealth Fighter
- FEATURES: Low Observability (Stealth), Supercruise, Diverterless Supersonic Intake (DSI)
- DEVELOPMENT: ADA (DRDO) + HAL. Focus on Private Sector involvement for prototype



PROJECT KUSHA (LR-SAM)

INDIA'S INDIGENOUS "S-400"
(Long Range Surface to Air Missile)

- RANGE: Up to 350 km
- TARGETS: Fighter Jets, Stealth Aircraft, Drones, Cruise Missiles
- SIGNIFICANCE: Comprehensive Air Defence Dome

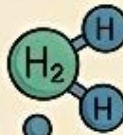
PREPARED BY: **ABHISHA AI**
by Abhishek Pathak

UPSC PRELIMS 2026: ENERGY & SUSTAINABILITY UPDATES

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1. NATIONAL GREEN HYDROGEN MISSION (NGHM)

💡 **Mission Phase (2026):**
Transition to Phase II - Commercial-scale integration.



💡 **Electrolyzer Manufacturing:**
Under SIGHT scheme, 3,000 MW annual capacity awarded. Units operational by late 2026.



💡 **Hard-to-Abate Sectors:**
5 pilot projects for Steel.

- **Focus:** Direct Reduced Iron (DRI) and blending in blast furnaces.



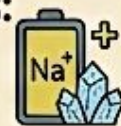
💡 **Key Fact:** Target 5 MMT Green Hydrogen production per annum by 2030.

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2. ADVANCED BATTERY ECOSYSTEM

💡 **Sodium-ion (Na-ion) Batteries:**

- **Advantages:** Abundant sodium (common salt), lower cost.
- **Early 2026:** First mass-produced Na-ion commercial EVs globally.
- **India Context:** Primary solution for stationary storage (grid backup) and low-cost EVs.



💡 **Solid-State Batteries (SSB):**

- **Mechanism:** Replaces liquid electrolyte with solid (ceramics/polymers).
- **Benefits:** Higher energy density (400-500 Wh/kg), faster charging, lower fire risk.
- **Status:** India prioritizing R&D for SSB.



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3. CARBON CAPTURE & UTILISATION (CCU)

💡 **The Five Testbeds:**
Approved by Min. of Science & Technology in Cement Sector



💡 **Why Cement?:**

CO₂ is a direct chemical process byproduct (calcination of limestone). Hard-to-abate.



💡 **CCU vs. CCS:**



- **CCU (Utilisation):** Converts captured CO₂ to synthetic fuels, urea, soda ash, concrete aggregates. (Circular Economy).
- **CCS (Storage):** Buries CO₂ geologically.

💡 **Economic Shield:** Testbeds help bypass EU's Carbon Border Adjustment Mechanism (CBAM).

